

Matching Compressor

User Guide

1. Introduction

Matching Compressor is a multi-knee feedforward compressor (up to three knees) that allows you to match the dynamic characteristics of one audio signal to another. In this guide, a knee means one segment of the processing curve, with its own threshold, ratio and knee width, not only the rounded transition around a threshold.

If you have a reference audio signal with your desired dynamic profile and another signal you want to shape similarly, the plugin can analyze both and determine compressor parameters that best match the reference.

Key features:

- Standard set of parameters: Threshold, Ratio, Knee Width, Gain, Attack, Release, along with Envelope Type and Stereo Processing mode
- High-pass filter (HPF) of the envelope input, with listen mode
- Support for ratios below 1:1, producing upward expansion above the threshold (downward expansion is not provided)
- Up to three knees
- Level distribution lock mode that keeps the level distribution unchanged when attack, release and HPF frequency are retuned
- Automatic matching of dynamic parameters
- Optional K-weighting when comparing level distributions
- Support for both audio files and input buses (main and sidechain) as sources for reference and destination audio
- Graphical representation of the processing curve

Level distribution lock mode becomes available once the destination audio has been analyzed.

Matching Compressor is distributed as a VST3 plugin for Windows and Linux. A MacOS version (VST3 and AU) is also available, but it is not signed or notarized; use it at your own risk.

The source code of the plugin is available at:

<https://github.com/justonem0reuser/MatchingCompressor>.

2. Working with matching compressor

2.1. Main window

2.1.1. Elements description

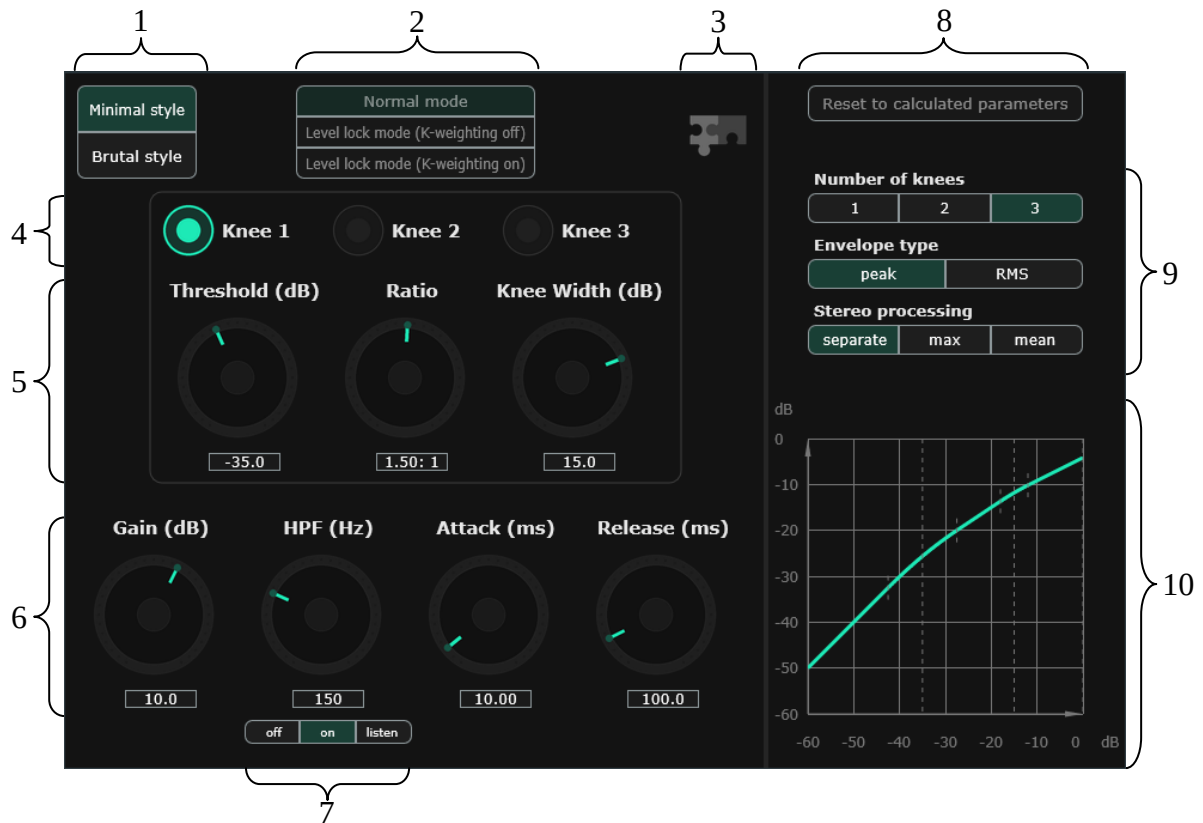


Fig. 1. Main window

The main window contains the following elements:

1. Style switching buttons – Change the visual style.
2. Mode switching buttons – Switch between the normal and the level distribution lock modes; the latter can be used with or without K-weighting (see section 2.1.4).
3. **Match/Learn** button – Opens the matching window, where matching against a reference signal, or learning from the destination signal alone, is started.
4. Knee selection buttons – Select the active knee (two or three radio buttons; hidden if only one knee is used).
5. Knee-related sliders:
 - **Threshold** – Sets the signal level at which downward compression or upward expansion begins.
 - **Ratio** – Determines how strongly the signal is reduced or increased once it crosses the threshold.

- **Knee Width** – Controls how gradually processing is applied around the threshold.

6. Chain-related sliders:

- **Gain** – Adjusts the overall output level after processing.
- **HPF** – Sets the cutoff frequency of the high-pass filter applied to the envelope input; the audio path is not filtered.
- **Attack** – Sets how quickly the processor responds after the signal exceeds the threshold.
- **Release** – Controls how quickly the processor returns to normal after the signal falls below the threshold.

7. HPF buttons (**off**, **on**, **listen**) – Turn the filter off or on, or switch to listen mode, in which the plugin outputs the filtered signal and no compression is applied.

8. **Reset to calculated parameters** button – Resets all the plugin-related buttons to their default or calculated values (disabled until matching is performed).

9. Plugin-related buttons:

- **Number of knees** – Sets the number of knees in the processing chain (1, 2 or 3).
- **Envelope type** – Selects whether detection is based on **peak** levels (fast, precise) or **RMS** levels (smoother, average-based).
- **Stereo processing** – Defines how stereo channels are processed (no effect on mono audio): independently (**separate**), linked by the louder channel (**max**), or linked by the average level (**mean**).

10. Plot – Displays the compressor curve and the fit quality.

2.1.2. Working with sliders and knees

Double-clicking a slider resets it to its default state.

Before the first matching operation, the default values are:

- **Threshold:** 0 dB
- **Knee Width:** 0 dB
- **Ratio:** 1:1
- **Gain:** 0 dB
- **HPF:** off

These correspond to no processing with a hard knee. Default **HPF** frequency, **Attack** and **Release** values are 100 Hz, 10 ms and 100 ms, respectively.

After matching, default values correspond to the calculated results (for **Threshold**, **Knee Width**, **Ratio**, **Gain**) or fixed values (for **HPF** frequency, **Attack** and **Release**).

If the **Ratio** is less than 1:1, expansion is applied instead of compression.

Selecting a knee using the radio buttons links the knee-related sliders to that knee.

If multiple knees are used, they cannot overlap and are kept in ascending order. The maximum threshold is 0 dB. Therefore, if the upper boundary of a knee (threshold + half of its width) reaches 0 dB, adding another knee becomes impossible (the corresponding option is disabled). Moreover, the available ranges of the **Threshold** and **Knee Width** sliders are limited by these constraints.

2.1.3. Plot

The plot uses a standard representation for dynamic processors: the X-axis shows the input level in dB, and the Y-axis shows the output level in dB.

The thick curve shows the current settings. The thin curve shows the parameters obtained during matching (if available).

Vertical dashed lines indicate each knee's threshold, and shorter dashed lines indicate knee boundaries.

After matching is performed or in the level distribution lock mode the quality indicator is also displayed above the plot:

- In the normal mode after matching “*mismatch against reference*” is displayed: it shows how the current compression and envelope parameters fit the level distribution of the reference audio.
- In the level distribution lock mode “*mismatch against fixed dynamics*” is displayed: it shows how well the new compression curve with the new HPF frequency (if the filter is on), attack and release values fits the level distribution obtained from the initial compression and envelope parameters.

The value displayed is the non-negative measure of mismatch: the closer it is to zero, the better the fit quality.

2.1.4. Working in the level distribution lock mode

After learning or matching is performed (see section 2.2), the level distribution lock mode becomes available. To enter it, click the **Level lock mode (K-weighting off)** or **Level lock mode (K-weighting on)** button. The two buttons differ in whether K-weighting is applied when level distributions are compared (see section 2.2.3).

In this mode, only **HPF** (if the filter is on), **Attack** and **Release** sliders are enabled; **Threshold**, **Ratio**, **Knee Width**, and **Gain**, along with **Number of knees**, **Envelope type**, **Stereo processing**, and HPF mode cannot be changed.

While using the level distribution lock mode, changing HPF frequency, attack or release does not change the audio fragment level distribution significantly: after each change, a re-estimation runs that calculates the best threshold(s), ratio(s), and gain to keep the resulting

change in the level distribution to a minimum. This mode can be used for facilitating the process of the HPF frequency, attack and release values tuning.

Note that each recalculation takes time up to several seconds. For this reason, the new values of HPF frequency, attack and release are not applied to the sound immediately; they are applied only together with the recalculated compression curve parameters. To monitor the process, see the text displayed above the plot: if “Calculating...” is shown, the process is not finished yet; if “*mismatch against fixed dynamics*” is shown, the new parameters are calculated and applied, along with the HPF frequency, attack and release set by the user.

It should be mentioned that in the level distribution lock mode knee widths are not calculated: they retain their values unless that would make neighboring knees overlap. Only in this case they are narrowed to prevent overlapping.

To return to normal mode, click the **Normal mode** button.

2.2. Matching window

To open the matching window, click the **Match/Learn** button.

2.2.1. Elements description



Fig. 2. Matching window

This window contains:

1. Reference and destination audio panel.
2. Matching settings.
3. **Match (Learn)** and **Cancel** buttons.

2.2.2. Setting reference and destination audio

Reference audio is the signal with desired dynamics. **Destination audio** is the signal to be processed. Both can be loaded from files or recorded from buses.

From files:

1. Click the **Open from file** button.
2. Select a file from a dialog.
3. A confirmation message will be displayed.

From buses:

The destination audio is read from the main bus, which is always connected to the processed track. The reference audio is read from the sidechain bus, which is usually disconnected: connect it to the required track in your DAW, otherwise the **Read from sidechain bus** button remains disabled.

1. Ensure that the required bus is connected via your DAW.
2. Stop playback.
3. Choose which of the audios (reference, destination or both) should be set from buses by clicking appropriate buttons: **Read from sidechain bus** and/or **Read from main bus**. The clicked buttons change their color.
4. Start playback from the desired point. The two buttons become disabled until stop.
5. Stop playback after the desired fragment.
6. A confirmation message will be displayed.

The **Match (Learn)** button becomes available only after the destination signal is set. If the reference signal is also set, the button's text is **Match**; otherwise, its text is **Learn**.

2.2.3. Matching settings and process

If the **K-weighting** option is on, the reference signal and the processed destination signal are K-weighted before their level distributions are compared. K-weighting affects only this comparison: neither the audio path nor the envelope is filtered by it. Hearing sensitivity depends on frequency: tones of equal amplitude but different frequencies are perceived as differing in loudness. K-weighting approximately accounts for this effect by filtering the signals. The levels are compared without block-based averaging, so, strictly speaking, they do not represent loudness itself; nevertheless, K-weighting brings them closer to how the human ear perceives audio material. In the level distribution lock mode, K-weighting is selected with the mode switching buttons (see section 2.1.4).

The **Knee type** parameter has two options. If the **Hard knee** option is selected, knee widths are not determined and are set to 0 dB instead. The **Soft knee** option means that knee widths are determined.

Number of knees, Envelope type, Stereo processing, HPF, HPF frequency, Attack, and Release are parameters that are fixed during the estimation process.

If the reference signal is set, the estimation process starts after clicking the **Match** button. Once it is finished, the matching window closes and the processing parameters are set:

- **Threshold, Ratio, Knee Width** for each knee, as well as the overall **Gain**, are set based on the calculation results.
- **Number of knees, Envelope type, Stereo processing, HPF (on/off), HPF frequency, Attack, and Release** are applied from the matching window.

If the reference signal is not set, all matching settings are unavailable and have no effect. After clicking the **Learn** button, the matching window closes and the level distribution lock mode becomes available.

3. Tips

- The processed audio will sound more similar to the reference if both have similar content. Matching a reference is meaningful only insofar as the fragments being compared are close in timbre and articulation.
- On sparse material, threshold and ratio can be poorly separated: different pairs of values give almost the same match. In this case, compression with different parameters can sound similar, but the ratio found can be far from the expected value or even hit its limit.
- If low frequencies dominate the envelope and cause pumping, turn on the **HPF**: the envelope will respond less to bass, while the audio path itself is not filtered. Listen mode lets you hear the signal the envelope is calculated from.
- Setting attack and release times to zero is not recommended for any compressor. The plugin does not attempt to estimate optimal attack and release values; they must be specified manually.
- Since matching involves optimization of a non-linear function, convergence to a global minimum is not guaranteed. Increasing the number of knees (especially when using the **Soft knee** option) does not always yield more accurate results.
- Using the **Hard knee** option allows you to reduce computation time compared to the **Soft knee** option with minimal loss of accuracy.